F(AI)^{2}R: Who Did What, and Who Checked?

Verifiable AI Provenance as an Executable Skill

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Abstract

F(AI)^{2}R is FAIR research with AI in the loop, twice: an AI-assisted *authoring* pass and a machine-readable *audit* pass over every artefact. AI systems now draft, refactor, and verify research artefacts, yet their contributions are rarely recorded in a form a later human or machine can audit. Building on the original F(AI)^{2}R experiment, we generalize its provenance model beyond scholarly writing into *aiprov*, a PROV-O extension covering any AI-in-the-loop artefact, and we package the method as an *executable skill* that an AI agent operates itself: setup asks the human operator for their ORCID iD, resolves their identity from the public registry, and scaffolds continuous integration that gates every push on graph conformance and publishes the current build of this very paper. The paper is its own case study. Every activity, claim, and source in its production is recorded in the repository's provenance graph under two invariants: no parentless claim, and verification rungs that only humans may grant.

Keywords: AI provenance; PROV-O; research integrity; agent skills; FAIR; human oversight

1 Introduction

F(AI)^{2}R is not an established term, and that is deliberate. It reads as FAIR (Wilkinson et al., 2016) with the AI factor squared, because AI enters research twice: once *authoring* an artefact, and once *auditing* it into a machine-readable record of who did what, when, and from which sources. This paper defines that double role precisely, generalizes it beyond scholarly writing, and demonstrates it on its own production.

AI agents now draft manuscripts, refactor code, sweep literature, and verify results. The debate about whether they should is loud and well documented (Dwivedi et al., 2023; van Dis et al., 2023); the record-keeping has not kept pace. A git commit stores *what* changed and a free-text author name; it does not record whether that author was a human or a model, at what cost the change was made, under which instructions, or from which sources. The work is real, but its context evaporates at commit boundaries. Existing responses are either policy (disclosure statements, authorship bans) or documentation artefacts written after the fact, such as model cards (Mitchell et al., 2019); neither is machine-checkable at the granularity of a single claim.

We treat this as a question of scientific ethics, and we state our position plainly. When AI co-writes, the ethical question is not "is that allowed?" It is: where does this come from, and can anyone check it? Integrity then becomes a property of the record, not of the author's assurances. Our stance is a measured one: not banning AI from research, and not trusting it blindly, but building the layer on which trust can be grounded. Provenance is that layer. A verification ladder says who may vouch for what, and its top rungs are reserved for humans. The incentive system makes this urgent rather than optional: publish-or-perish pays for counted output, and hypercompetition selects for whatever maximizes the count (Edwards and Roy, 2017). Generative AI collapses the marginal cost of exactly the thing the metric counts, onto a review system already strained by output growth (Hanson et al., 2024). Section 3 describes the predictable result. Appeals to ethics alone do not bend that curve. Infrastructure that makes verifiable work cheaper than unverifiable work can. The most expensive mistake would be to scale speed without traceability.

This is a fast paper standing on slow work. The ideas were sharpened across earlier experiments: *Obscurity-Is-Dead* (Krebs, n.d.b) established transcript-as-artifact and per-claim verification labels while measuring how large language models compress reverse-engineering effort, and the original F(AI)^{2}R experiment (Krebs, n.d.a) formalized the two-pass model and the no-parentless-claim rule on a single scholarly paper, bound to paper writing. Several strands of that programme advance in parallel under one operator: this method, this paper, and a system-integration fork of the shepard data-management platform (Krebs, n.d.c). That is precisely why per-activity accounting matters: parallelism without provenance is how contributions and errors alike become untraceable.

The paper makes three contributions. First, *aiprov*, a domain-agnostic generalization of the F(AI)^{2}R vocabulary over PROV-O (Lebo et al., 2013): agents, passes, claims, full inference telemetry, and a verification ladder with human-only rungs (Section 4). Second, the method packaged as an *executable skill* that an AI agent operates itself, from an ORCID-first setup through continuous integration that gates every push on graph conformance and publishes the current build of this very paper (Section 5). Third, a meta-experiment: the paper is its own case study, and its provenance graph, including mishaps and mid-experiment method changes, is the evaluation object (Section 6). Sections 2 and 3 position the work and motivate it; Sections 7 and 8 discuss limits and close the loop.

2 Background and Related Work

Provenance models. W3C PROV-O (Lebo et al., 2013) is the standard vocabulary for expressing who generated what, through which activity; it consolidated the lineage begun by the Open Provenance Model (Moreau et al., 2011), and PAV refined it for authoring and versioning (Ciccarese et al., 2013). Carina Haupt's overview of provenance use cases documents the practice inside DLR, including monitoring software development processes with provenance data (Haupt, 2022). aiprov extends this stack rather than forking it: every class subclasses a `prov:` term, so generic PROV tooling keeps working.

Contributor attribution. CRediT names who contributed what kind of work (Allen et al., 2019) and is now a NISO standard (NISO, 2022), but it operates at manuscript granularity and admits only humans. aiprov records attribution per activity and per claim, and its agent model includes AI systems and deterministic tools alongside people.

AI transparency artefacts. Model cards (Mitchell et al., 2019) and datasheets (Gebru et al., 2021) document the *model* or the *dataset*. What they do not capture is the collaboration that produced a specific artefact: which prompts, which sessions, which verification steps. RO-Crate (Soiland-Reyes et al., 2022) is complementary packaging; an aiprov graph can travel inside a crate. Closest to this work, PROV-AGENT extends W3C PROV to capture agent interactions, prompts, responses, and decisions, in agentic workflows in near real time (Souza et al., 2025); it shares the substrate and the agent-centric telemetry, and differs in what this paper adds on top: verification rungs reserved for humans, packaging as an executable skill, and the method applied to its own production.

The trust gap. The large language models that now write research prose descend from the transformer architecture (Vaswani et al., 2017), and they fabricate plausible citations (Alkaissi and McFarlane, 2023): existence of a reference is machine-checkable, support for a claim is a judgement, and the two must not be conflated. The ladder of Section 4 keeps them apart and names the grantor of every judgement.

Neighboring lines beyond provenance vocabularies. Claims as first-class citizens predate this work: nanopublications make a single assertion with its provenance a citable object (Groth et al., 2010), and micropublications model claims, evidence, and arguments formally (Clark et al., 2014). aiprov inherits that stance and adds what the AI era demands: the agent class behind every activity, and a verification ladder whose grantor is explicit and whose top rungs are human-only. On the artefact side, C2PA Content Credentials sign machine-readable provenance manifests for media, including AI-generated content (Coalition for Content Provenance and Authenticity, 2025), and in-toto binds software artefacts to their supply-chain steps with signed attestations (Torres-Arias et al., 2019). Both certify what a pipeline did to bytes; aiprov records how humans and AI divided the research work and who vouched for which claim, the layer the dual-witness ask of Section 7 would connect to this signing infrastructure. Manubot demonstrated CI-built manuscripts with citation-by-identifier years earlier (Himmelstein et al., 2019). These four lines entered the record through the review that requested them: the original novelty sweep, two registry queries, missed all of them, the bounded-novelty mechanism working as designed, a narrow recorded horizon visibly widened.

Lineage and neighbors. This work has three roots. The first is AI-assisted research practice: *Obscurity-Is-Dead* (Krebs, n.d.b) introduced transcript-as-artifact and the verification labels (`repo-vendored`, `lit-read`, `unverified-external`) that the present ladder canonicalizes, and $\mathrm{F(AI)^2R}$ (Krebs, n.d.a) formalized the two-pass model and no-parentless-claim. The second is engineering provenance: adjacent DLR work extends PROV-O with uncertainty quantification for traceability in engineering systems (Valente et al., 2026); the axes are complementary, since aiprov records who and how, while uncertainty-aware provenance records how confident, and a combined profile is future work (Section 8). The argument shape itself is established in research software engineering: better architecture makes better software makes better research (Druskat et al., 2025); aiprov extends that chain from the software to the record of its making. The third root is industrial semantic modeling: the author co-authored the Asset Administration Shell (AAS) Part 1 specification (Plattform Industrie 4.0, 2022), continued as IDTA-01001 (Industrial Digital Twin Association (IDTA), n.d.). Where partners exchange information across a value chain, semantics must be formal and machine-readable or interoperability fails. aiprov applies that discipline to a new exchange relationship: the one between humans, AI agents, and the future auditors of their joint work. The Helmholtz Metadata Collaboration (Helmholtz Association, n.d.; Helmholtz Metadata Collaboration, 2025) provides the community context: a provenance graph is metadata about *work*, extending FAIR metadata practice from data description to collaboration description.

3 The Dilution Crisis

Scientific output has been growing faster than the community's capacity to review it (Hanson et al., 2024), and paper mills industrialized fabrication well before language models (Heck et al., 2021). What changed is the cost curve. *Obscurity-Is-Dead* measured how language models compress the effort gap for reverse-engineering proprietary devices (Krebs, n.d.b); the same compression applies to producing a plausible paper. GPT-fabricated papers are already indexed and ranked by scholarly search engines (Haider et al., 2024).

The demand side is the incentive system. Publish-or-perish pays for counted output, and hypercompetition selects for whatever maximizes the count (Edwards and Roy, 2017). Language models did not create that pressure; they gave it a zero-cost supply channel. Dilution is misaligned incentives meeting free volume.

The consequence is that finding good sources is now the bottleneck of research. Retrieval trusts proxies: indexed, cited, fluent. Generation has learned to imitate all three. Language models fabricate plausible references and make citation errors at documented rates (Walters and Wilder, 2023; Alkaissi and McFarlane, 2023); once such a reference is indexed, nothing in today's retrieval chain stops it from being cited onward, and models trained on the polluted pool degrade further (Shumailov et al., 2024), so dilution compounds. The parallelism that lets one operator advance several strands equally industrializes pollution (Section 7); per-activity accounting is what tells the two apart. Research infrastructures already treat metadata quality as their central lever (Helmholtz Metadata Collaboration, 2025), and the deployed remedy is graph-shaped: the Helmholtz Knowledge Graph harvests metadata from siloed infrastructures precisely to drive better metadata practice (Bröder et al., 2025). The diluted literature is, in our words, a data swamp one level up, and it admits the same remedy.

When text is cheap, the scarce good is an original idea whose support is checkable. We treat originality as a provenance property: a novelty claim carries its recorded search horizon, that is, what was searched, where, and when. Concretely, the tooling's registry sweeps (Crossref, OpenAlex, arXiv, DataCite) are logged activities with their query terms, so "no prior work found" stops being an unfalsifiable assertion and becomes a bounded one: no prior work found *within this recorded horizon*, which a sceptical reader can re-run, widen, and possibly refute. The literature sweeps behind this paper are logged exactly so. Every claim must then be supported, or refuted, by facts, and both outcomes leave a record.

aiprov's answer makes claims first-class. Each assertion carries `prov:wasGeneratedBy`, `prov:wasAttributedTo`, and a ladder rung that separates "the reference exists" (machine-checkable) from "the content supports the claim" (machine-judged at most) from "a human checked" (human-only). Refutation is explicit: `aiprov:contradicts` records a claim refuted by facts as refuted, instead of silently deleting it. This is supply-side transparency. Artefacts carry a pedigree that fabrication cannot cheaply forge: DOIs that resolve, content hashes, commits, and registry-resolved identities. Filtering the pool can then shift from fluency heuristics to verifiable metadata.

4 The aiprov Method

Model. aiprov extends PROV-O with three agent classes (`HumanAgent`, `AIAgent`, `ToolAgent`), four activity classes (`AuthoringPass`, `AuditPass`, `Build`, `Repair`), and entity classes for artefacts, claims, sources, transcripts, and prompts, where prompts are treated as source code and bound to activities as plans (Figure 1). Humans carry ORCID and affiliation; AI agents carry model, version, provider, endpoint, context window, and knowledge cutoff; tool agents cover deterministic actors such as CI jobs. Treating prompts as plans deserves a word: a prompt is to an AI activity what a script is to a batch job, the executable specification of intent, so it is hashed and bound with `prov:hadPlan` rather than paraphrased into a comment. An auditor who wants to know *why* an artefact came out as it did reads the plan and the transcript; neither is recoverable after the fact if not captured when the activity runs.

Telemetry. Activities carry session and request identifiers, token counts by class (input, output, cache, reasoning), sampling parameters, cost, energy, and the tools invoked. One rule governs all of it: omit, don't estimate. Absent telemetry is recorded as absent; invented precision is worse than a visible gap. Generated artefacts receive sha256 content hashes at logging time, and prompt files receive prompt hashes.

Invariant I: no parentless claim. Every `aiprov:Claim` needs a generating activity, an attributed agent, and a verification state. The conformance validator treats violations as hard failures and is suitable as a pre-commit hook or CI gate.

Invariant II: the verification ladder. Rungs answer who checked what (Figure 2, Table 1). Promotions must strictly climb and are themselves logged as audit activities, so every rung a claim or source holds has a recorded grantor. Disagreement is first-class: an agent may *refuse* a promotion, which changes no state but is logged as an audit activity carrying the refused rung, so "checked and not convinced" is distinguishable from "never checked". Rungs 5–6 are human-only; an AI-granted promotion to them is a validator hard failure. Two rungs deserve comment. Vendoring (rung 4) is not a verification statement: an unread vendored copy proves nothing beyond `reference-resolved`. Its role is instrumental, the access gate that makes human verification feasible and durable. Operationally, the agent must hand the human the evidence: `promote` refuses the human-only rungs unless the source is vendored (path and hash recorded) or carries a clear access link, and prints the review material with every request; a human confirmation against link-only material is flagged, since the audit target remains mutable. At the top, `human-read` subsumes `human-confirmed`: it asserts full-text familiarity *and* confirmation, because a name alone would mislead, as reading does not entail checking. Both refinements were operator-contested mid-experiment (Section 6). The ladder itself has provenance: it canonicalizes labels field-tested in *Obscurity-Is-Dead* (Krebs, n.d.b) and hardened in F(AI)^{2}R (Krebs, n.d.a), and legacy names remain machine-readable aliases, so that ancestor graphs can consolidate onto this ladder without rewriting history; the aliases are in place, though the consolidation has not yet been exercised against the ancestor repositories' graphs.

Commit binding. An activity can only bind a commit that already exists, and a commit cannot contain the graph entry describing itself. The working discipline that follows: commit the artefacts, log the activity with that commit's hash, then commit the graph update separately. The graph never claims a hash it could not have known; the cost is one extra commit per logged round. What the two commits buy is tamper evidence in both directions. The graph entry names a commit whose tree contains the artefacts it describes, content-hashed; the graph itself lives in later commits of the same history. Backdating a record would require rewriting published git history, and quietly swapping an artefact would break its recorded hash: either falsification is possible, but neither is silent; that asymmetry (honest work is cheap, dishonest work is loud) is the design goal throughout.

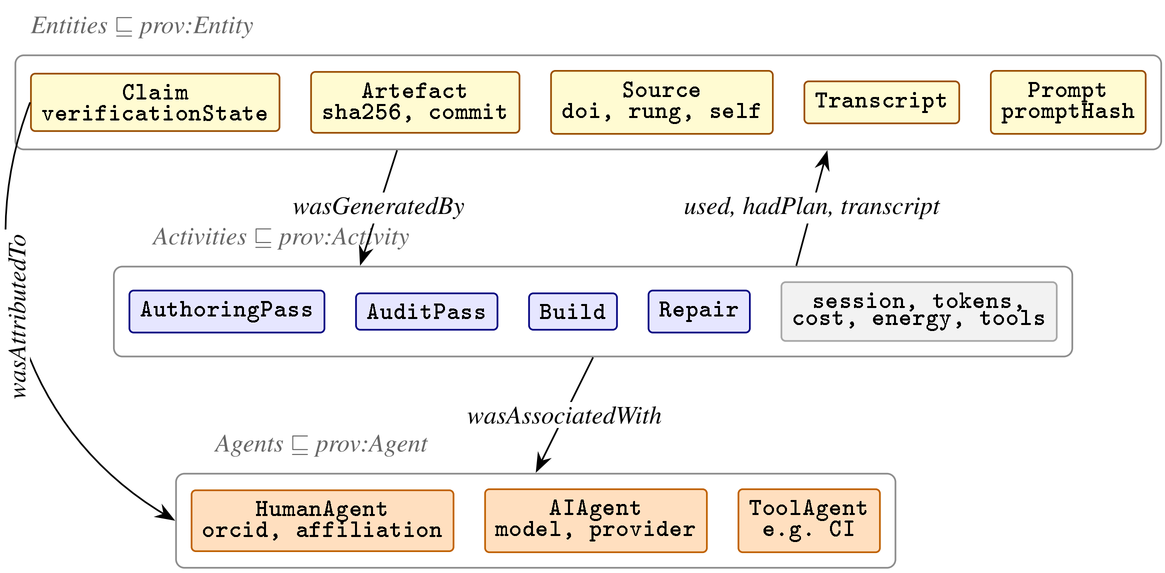

Figure 1: Core aiprov vocabulary as an extension of PROV-O. Every class subclasses a `prov:` term; tiny annotations name characteristic attributes. Claims point to their generating activity, their attributed agent, and a verification state; activities carry inference telemetry and link prompts (as plans) and transcripts.

The validator, exhaustively. Conformance is seven concrete checks, each a SPARQL query over the graph, and naming them makes the invariants operational rather than aspirational. Three enforce Invariant I: no claim without a generating activity, no claim without an attributed agent, and, as a warning, no agentless activity. Two enforce Invariant II: no human-only rung granted by an AI agent, and no human-only rung without a recorded promotion activity naming a human grantor, a warning that flags a rung edited directly into the graph without an audit trail. Two enforce the access gate: no source above `needs-research` without a resolvable DOI, URL, or vendored copy, and a warning, rather than a failure, for human-verified sources whose audit target is link-only and therefore mutable. Four checks fail hard with a nonzero exit code; that exit code is the entire integration contract, which is why the same command serves unchanged as a pre-commit hook, a CI gate, and an auditor's first step.

Auditability operations. `report` aggregates tokens, cost, and rung distributions; `extract` computes the backward closure of everything that contributed to chosen entities; `dashboard` renders the graph as a self-contained offline ledger. Sources enter through open registries (Crossref, OpenAlex, arXiv, DataCite): a search command sweeps them, and registration with DOI verification promotes a source to `reference-resolved` only if the registry actually resolves it.

5 The Executable Skill

A method that lives in a PDF of guidelines depends on humans remembering to follow it. We package aiprov instead as a *skill*: a versioned bundle of instructions and tools that an AI agent loads and operates. The method's rules become the agent's operating rules: log your own activities, record claims at `ai-confirmed` at most, never fabricate telemetry, and never grant a human-only rung.

ORCID-first setup. Setup asks the human operator exactly one question: their ORCID iD. A single command then seeds the graph, resolves the operator's name and current affiliation from the

Table 1: The verification ladder. Positions are machine-readable (`aiprov:ladderPosition`); promotions must strictly increase them and are themselves logged as audit activities, so every rung a claim or source holds has a recorded grantor.

Pos	Rung	What it asserts	Who may grant
0	`unverified`	Recorded; nothing checked yet by anyone. Default state at creation.	(default)
1	`needs-research`	A check was attempted or demanded and did *not* succeed (e.g. a DOI failed to resolve). Must not be cited or relied upon until re-verified.	any agent
2	`reference-resolved`	The reference *exists*: its DOI/URL resolved in a bibliographic registry and metadata was captured. Says nothing about content.	any agent (tool-checkable)
3	`ai-confirmed`	An AI checked that the source *content* supports the associated claim. A machine judgement, the highest rung an AI may grant.	AI or human
4	`source-vendored`	A copy of the source is preserved inside the repository (content-hashed), immune to link rot and silent revision. Of no evidential value by itself: this rung is the *access gate* to human verification; the agent hands the human the evidence, and tooling refuses rungs 5–6 unless the source is vendored or carries a clear access link (DOI/URL).	any agent (hash-checkable)
5	`human-confirmed`	A *human* spot-checked that the source supports the claim.	human only
6	`human-read`	A *human* has read the source in full *and*, with that full context, confirms the claim is supported, subsuming rung 5's confirmation with deeper familiarity. Reading without confirming is not a rung. Top of the ladder.	human only

public registry (never guessed; if resolution fails, only the bare iD is recorded), derives the instance IRI base from the git remote, scaffolds continuous integration, and lays down a compilable chapter-per-file paper skeleton whose author block is already filled from the resolved identity. Later commands auto-detect the base from the graph, so configuration never needs repeating (Figure 4).

CI as the deterministic build agent. Every push validates the graph server-side, renders the dashboard, regenerates the AI-transparency disclosure, and compiles the paper to PDF, uploading dashboard and PDF as build artefacts. The current version of the paper is always available, and a conformance violation blocks the build. The CI service itself is registered as a `ToolAgent`, and its green runs are logged `Build` activities carrying artefact digests. Version tags extend the same discipline to releases: a `v*` tag publishes a conformance-gated release whose assets are the PDF together with the graph it was built from, the metrics snapshot, the packaged skill, and their checksums, so a release is the artefact plus its record.

Live preview as oversight affordance. On request, the agent republishes the current typeset build, together with the per-section planning skeletons, to a stable preview URL after every editing round (Figure 5, right panel). Effective oversight of a fast-moving AI collaborator needs a continuously current view of the deliverable; the EU AI Act's human-oversight expectation (European Parliament and Council of the European Union, 2024) becomes a user-interface property. The preview also renders the source-verification queue, whose buttons open prefilled repository issues; a workflow verifies the issue author against a registered-operator mapping and transcribes the authenticated grant or refusal into the graph, so the operator walks the human rungs without touching a checkout: the judgement

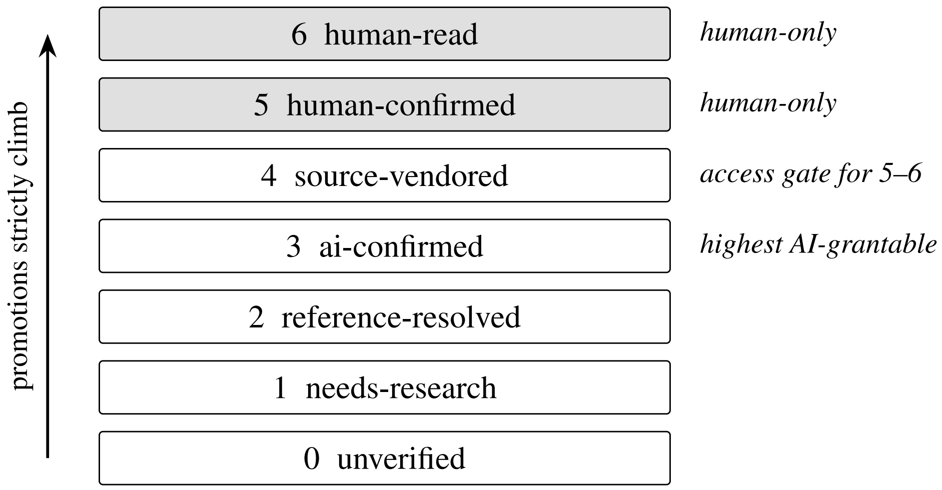

Figure 2: The verification ladder. Every promotion is logged as an audit activity; rungs 5–6 may never be granted by an AI agent, and the validator treats such grants as hard failures.

stays the human's, only its transcription is automated.

Structure before prose. The agent first lays down each section's principal structure, its argument flow, evidence, and figure and citation plan, as a reviewable skeleton; prose is a separate later pass. The operator can contest structure cheaply before any prose exists to defend it, and both passes are logged, so the audit trail shows which structural decisions preceded which text. Two rungs of the ladder were redesigned exactly this way during this experiment.

Regulatory fit by construction. The EU AI Act transparency statement is derived from the graph on every build (Section 7); disclosure is a build product, not an authoring chore, and adopters inherit the mechanism.

Worked example: analysing experimental data. This example is a prospective design, not yet an implemented deployment. The workflow is not specific to writing papers; consider a study like the comparison of power and amplitude control in continuous ultrasonic welding of unidirectional carbon-fibre-reinforced polymers (CFRPs) (Janek et al., 2025), where welding trials produce in-situ process data, scans, mechanical test results, and micrographs, and the outcome is comparative claims about two control strategies. Setup is identical: one ORCID question seeds the graph and registers the experimenters; the rig's data acquisition and the analysis scripts join as `ToolAgents`, the AI assistant as an `AIAgent`. Each raw record (a weld's power trace, a scan, a test protocol) is registered as a `Source` and vendored with its content hash, so the access gate of rung 4 holds the evidence for every downstream claim as immutable bytes. Each analysis run is a logged `Build` activity that `used` the raw sources and `generated` the derived tables and figures, hashes included; CI reruns the analysis, so a figure that cannot be regenerated from the recorded inputs fails the build. AI-assisted interpretation is an `AuthoringPass` with telemetry, and every finding (say, that the two control modes yield equivalent weld quality on a given layup) is a claim generated by the analysis activity and attributed to its agent. The ladder then works exactly as in Section 4: the AI may cross-check a claim against the derived data and grant `ai-confirmed`; the experimenter spot-checks the mechanical-test table for `human-confirmed`, with the tooling handing over the vendored data on request; the comparative conclusions that carry the paper deserve `human-read`. The disclosure statement and the report come for free, and once the shepard integration planned in Section 8 lands, raw data and provenance will share

```
$ provlog.py validate
[ OK ] parentless claims: 0
[ OK ] unattributed claims: 0
[ OK ] human-only rungs granted by AI agents: 0
    (...four further checks OK...)        exit 0

# an AI agent tries to grant a human-only rung
$ provlog.py promote --id demo-claim \
    --to human-confirmed --agent claude-code
REFUSED: rung 'human-confirmed' is human-only;
  agent:claude-code is not a registered
  HumanAgent. An AI must never grant this rung.
                                          exit 1
# a parentless claim spliced into the graph
[FAIL] parentless claims: 1
    -> .../prov/claim/rogue               exit 1
# access gate: human rung, source not in hand
$ provlog.py promote --id demo-src \
    --to human-confirmed --agent florian-krebs
REFUSED: 'human-confirmed' requires the source
  in hand -- vendor a copy first (promote
  --to source-vendored --file <path>) or
  record a DOI/URL for it                 exit 1
# vendor the copy, then the gate hands it over
$ provlog.py promote --id demo-src \
    --to source-vendored --file protocol.pdf
demo-src: reference-resolved -> source-vendored
$ provlog.py promote --id demo-src \
    --to human-confirmed --agent florian-krebs
review material: vendored copy at protocol.pdf
demo-src: source-vendored -> human-confirmed
  (logged as AuditPass, agent:florian-krebs)
                                          exit 0
```

Figure 3: Success and failure, captured from the tooling running against a sandbox copy of this paper's graph: a clean validation, the two invariant refusals (human-only rung attempted by an AI; parentless claim), and the rung-4 access gate first blocking a human promotion and then, once a copy is vendored, handing the review material to the human. Output verbatim; long lines re-wrapped and exit codes annotated.

one backbone.

Auditability at the command line. Auditability here is not an abstract property but a concrete session: an auditor clones the repository and runs the same tooling the authors ran. Three checks need no trust in anyone's honesty. Re-running the validator replays every invariant against the graph as it stands; re-hashing a vendored file or a committed artefact and comparing against the recorded digest detects any silent swap; re-resolving a recorded DOI confirms the citation metadata still matches the registry. Two further checks read rather than execute: the activity log tells the auditor *who* granted each rung and under which commit, and the transcript shows the dialogue in which a contested decision was made. The tooling is built so that success is quiet and failure is loud, Figure 3: a clean graph prints one `OK` line per invariant and exits zero, suitable as a CI gate; a violation names the offending node and exits nonzero; a refused operation says why it was refused and what would make it legitimate. The last property matters most for the human rungs: when the operator asks for `human-confirmed`, the gate either blocks for lack of access material or prints exactly where the review copy sits, so the human step starts with the evidence in hand rather than a search for it.

Versioning the method. The skill tree is the repository: snapshots are immutable, changes are logged, and every activity binds its commit, so the meta-experiment can cite the exact method version in force at any point. The same property makes the skill the transferable unit; the bundle that wrote this paper is the intended integration payload for the shepard fork of Section 8. Conversations are part of the record too: a transcript exporter renders the session to a committed file after every turn, and activities link to it, so the dialogue that shaped an artefact is as retrievable as the artefact.

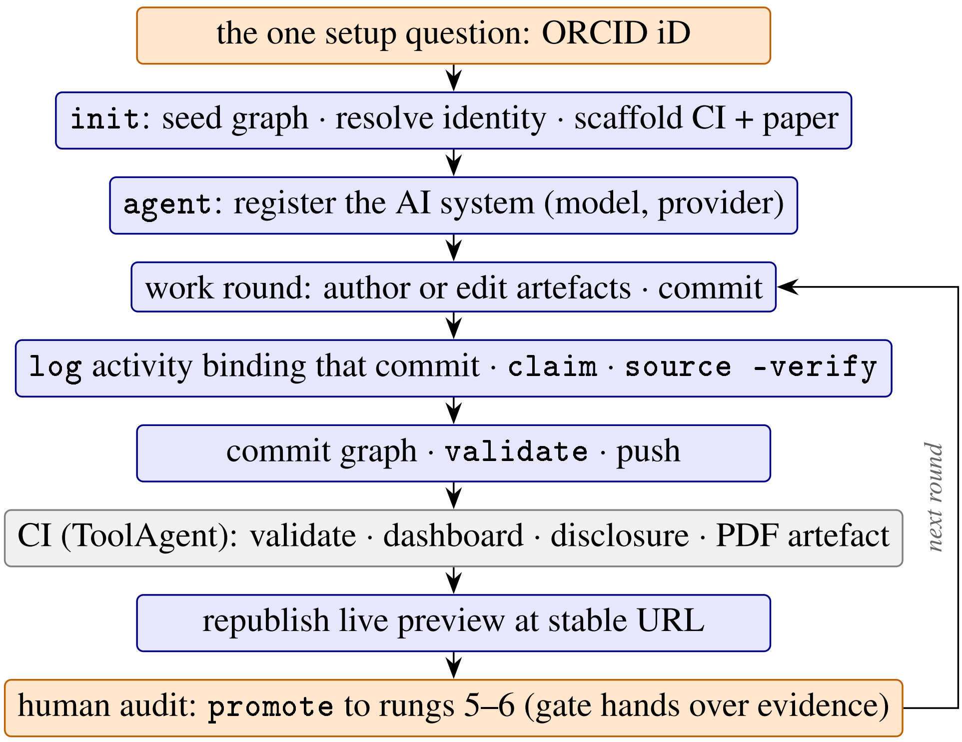

Figure 4: The skill's operating loop from the single setup question to the continuously published build. Orange steps involve the human operator; the grey step runs deterministically in CI; everything else is the AI agent operating the method, two commits per logged round (Section 4).

6 Meta-Experiment: This Paper as Case Study

Setup. The repository was bootstrapped by an AI agent operating the skill, in one working session directed by the human operator. Three agents are registered: one human (identity resolved from ORCID), one AI system, and the CI service as a tool agent. Every round of work follows the two-commit pattern of Section 4, so each logged activity binds a real commit. Figure 5 shows the oversight surface: the continuously rebuilt draft beside its source-verification queue, the loop of Section 5 in operation. Scope, before any number: this is a demonstration, not a study; one operator, one domain, one session. The quantities below characterize this instance of an idea in use (Section 8 lists the trials it owes).

The graph as evaluation object. Table 2 reports the state of the graph at the time of writing, and Figure 6 shows the ledger view rendered from the same file. Token counts were initially absent: the harness (the agent's execution environment) exposes no live usage to the agent, and under omit-don't-estimate nothing was recorded. The operator's challenge ("why is this not reported?") prompted a check of the session record, which turned out to carry the provider's per-request usage blocks; the totals in the table are aggregated from them, reported not estimated. Cost followed one step behind: the record carries no priced entries, so cost was first omitted; the operator then directed pricing the recorded tokens against the provider's published price list at time of writing. The resulting figure is *computed, not provider-reported*, and the graph records it exactly so, price basis in the activity's label. All eight recorded claims sit at `ai-confirmed`, the ceiling an AI may grant; promotion beyond waits for the human operator, with the evidence handed over on request.

Worked examples. Invariant enforcement was verified by test-run before any claim entered the

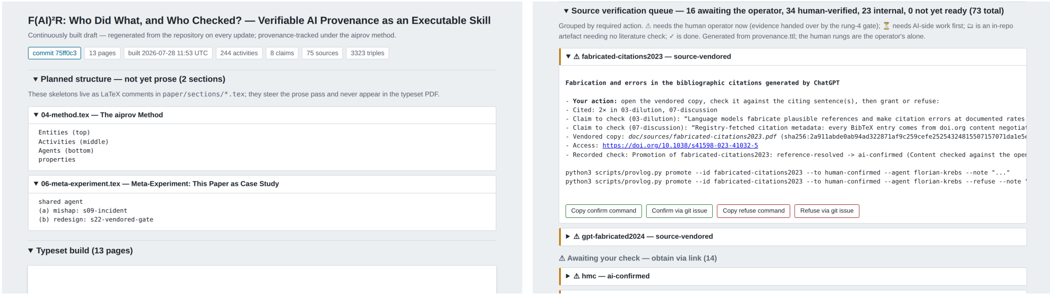

Figure 5: The continuously rebuilt preview of this paper (screenshot, 2026-07-28), republished to a stable URL after every editing round. Left: commit, build time, and live graph statistics above the typeset pages. Right: the source-verification queue at the bottom of the same page; each entry shows the citing sentence to check, the recorded evidence, the access link, and buttons that stage the exact grant or refuse command or open a prefilled repository issue, which a workflow transcribes into the graph after verifying the issue author against the registered-operator mapping. The judgement stays the operator's; only its transcription is automated.

graph: an injected parentless claim failed validation, and a promotion of a claim to `human-confirmed` attempted by the AI agent was refused, both captured in Figure 3. The bibliography was built through the method: sources found by registry sweeps, DOI-verified at registration, fetched as BibTeX from doi.org content negotiation, and cited only once ladder-backed. The paper's intellectual ancestors are themselves registered sources, each fetched or URL-verified first. Registration, however, only proves a reference *exists*; an audit pass flagged exactly this as the record's weakest point, and content passes closed it: for every cited source the AI fetched the actual content (full text or slides where openly accessible, abstracts otherwise) and checked it against the specific citing sentence, recording the supporting quotation and the checked depth in the promotion note; every cited source was checked at the time of writing. Negative outcomes stayed recorded rather than smoothed over: one publisher's bot-wall deferred a check until an open index carried the abstract, the operator-owned repositories were left to the operator's own rungs, and three citation overreaches surfaced by the checks were repaired in the text before any promotion.

Incidents as first-class records. The experiment produced four incident types, and none needed special vocabulary; each is an ordinary activity whose class carries its nature (Figure 7). (i) An agent mishap: while creating template stubs, a stale working directory led the agent to overwrite two committed manuscript sections. Version control surfaced the unintended diff immediately, restoration was a one-command revert, and the self-report is an `AuditPass` naming what happened, the tools involved, and the restoring commit. (ii) An infrastructure failure: a registry served malformed BibTeX, which broke the CI build; the diagnosis and fix live in commits, and the first green run is a logged `Build` activity by the CI tool agent carrying artefact digests. (iii) Blocked access: a publisher returned 403 to automated requests, and the fallback to the canonical page is recorded in the registering activity's label rather than silently swallowed. (iv) Method contestation: the operator challenged the semantics of two ladder rungs; the redesigns are ordinary authoring activities whose generated artefacts are the new tool and schema versions, with the enforcement claim verified by test-run. The pattern generalizes: erasing an incident would require rewriting git history *and* the graph, both visible acts, so honesty is the cheap path. The same property that makes contributions attributable makes errors reversible.

Division of labour as data. The record makes the division of work between operator, AI agent,

Table 2: Provenance-graph state of this paper's production at the time of writing (from `provlog.py report` and the repository; regenerated before submission). Token totals are aggregated from the provider-reported usage blocks in the session record; the cost is computed from those tokens and the published price list at time of writing (all cache writes in the record are 1-hour-TTL), not reported by the provider, and is recorded in the graph as computed.

Metric	Value
Triples	3 323
Logged activities	244
Claims (all `ai-confirmed`)	8
Sources on the ladder	75
thereof at `ai-confirmed` or above	73
thereof marked self-citation	5
Registered agents (human / AI / tool)	1 / 1 / 1
Commits	222
Transcript	1 session, 3 284 turns
Output tokens	2 551 954
Input tokens (uncached)	17 098
Cache tokens (read / write)	860.4 M / 32.6 M
Cost (computed, not reported)†	≈ USD 1641
same tokens at Opus 5 / Sonnet 5 rates†	≈ USD 820 / 328

†Basis: the provider's published API list prices (Anthropic, 2026), per million tokens: input $10, output $50, cache read $1, 1 h cache write $20; all cache writes 1 h-TTL. The repricing holds the recorded token volume fixed and only swaps the price list (Sonnet 5 at introductory rates); it compares price bases, not runs, since models tokenize and behave differently.

and infrastructure countable (Table 3). The class-by-agent counts are exact, machine-readable queries over the graph; attributing *triggers*, what prompted each activity, still requires reading the session log, and that reading is consistent: the operator's direction messages carry scope, lineage, people, and every contested semantic decision (both ladder redesigns, the cost policy, the license choice, the page budget), while the AI executed all authoring, audit, and repair activities and CI contributed the deterministic builds. Rungs above the AI ceiling remain human by design: the operator granted the first `human-confirmed` against a vendored statute text the gate handed over, confirmed the five self-cited sources they authored, the class where the human's judgement is uniquely authoritative, and then walked the queue through the issue path of Section 5, 34 human rungs granted at the time of writing. Authentication splits those grants honestly: 23 promotions carry GitHub-authenticated authorship through the issue path, itself derived mid-experiment, while 13 were transcribed in session from the operator's instruction, checkable only against the committed transcript; 29 of the human-verified sources remain link-only audit targets, the validator's warning kept visible. The remaining sources are deliberately kept at their machine rungs so every rung was exercised over the record's history (the final distribution occupies the upper four); the operator reports having checked all sources and claims, and the ladder records which of those checks they chose to formalize as grants. The hypothesis that AI absorbs tedium while humans connect ideas and people (Section 7) is thereby testable from the committed log and transcript, for this one experiment. One measurement pitfall became a method fix: the session format returns tool results under the *user* role, so a naive role count inflates the human by an order of magnitude (595 of 663 user-role messages were tool results). The transcript exporter now labels Human, Assistant, Tool result, and Hook distinctly; the same unambiguous role labeling is worth asking of provider session

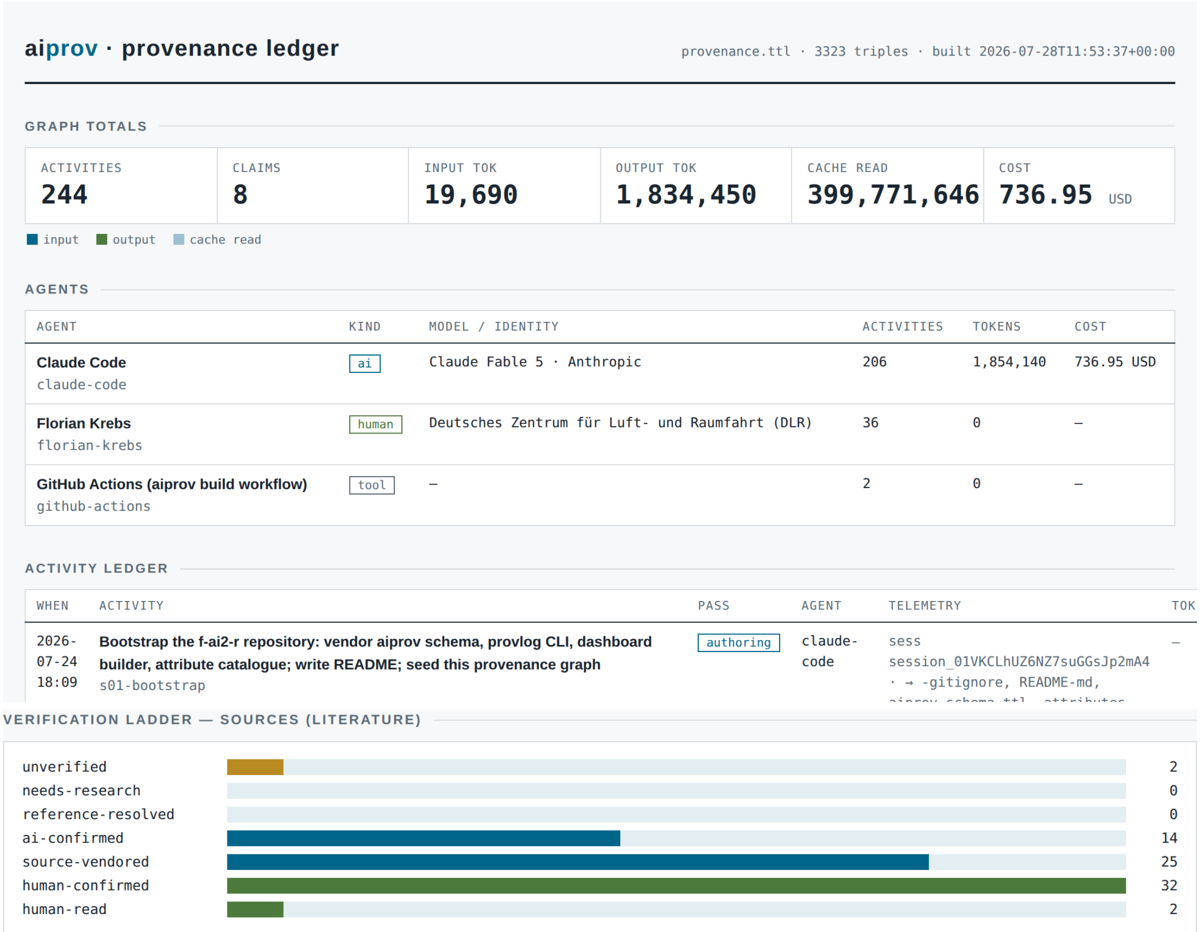

Figure 6: The dashboard rendered from `provenance.ttl` (screenshot, 2026-07-28). Top: header, graph totals, agents, and the start of the activity ledger; the token and cost tiles sum per-activity records (two aggregate backfills, roughly half the computed cost), so the session-record aggregates of Table 2 are the authoritative figures. Bottom: the literature rung distribution, most sources operator-verified, the top rung populated. The full page is one self-contained HTML file with an interactive graph view.

formats generally, in the same breath as provider-reported cost.

Leverage as data. The same record measures what the multiplication bought, without estimating any counterfactual. The operator typed 2 112 words of direction in total, median 7 words per message; active session time, clustering request timestamps and counting gaps under thirty minutes, is about 12.7 hours inside a wall-clock span of 90 hours. Against that input stand roughly 9 300 words of typeset prose, about 2 800 lines of tooling, 75 ladder-backed sources, and the graph itself. What a solo baseline would have cost is deliberately not estimated, omit-don't-estimate applies to counterfactuals too; what the record does show is *where* the freed attention went: nearly every typed word carries scope, semantics, lineage, or people, not prose. The gain is less the saved time than this: the human's whole contribution could be judgement. Every derived figure above follows the skill's derivation rule: its methodology is stated with the number, word counts strip LaTeX markup, direction counts exclude tool results, hook messages, skill loads, and pasted documents over five hundred words, and the thresholds ride in the measuring activity's label, so a sceptic can recompute or contest any of them.

What the auditor can and cannot verify. Hashes, commits, resolving DOIs, rung grants, and the validator's verdicts are checkable by anyone with the repository. The honesty of self-reported telemetry

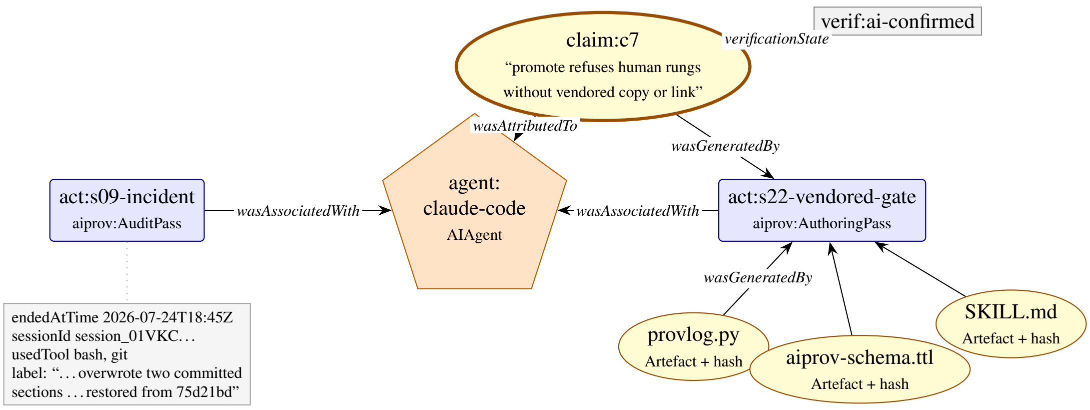

Figure 7: Two incidents from this experiment, exactly as the graph represents them (node contents abridged from real triples). Left, a mishap: the agent overwrote two committed manuscript sections; the self-report is an `AuditPass` carrying timestamp, session, tools, and the restoring commit in its label. Right, a contested design: the operator challenged the `source-vendored` rung's semantics; the redesign activity generated new tool and schema versions (content-hashed artefacts), and its enforcement claim sits at `ai-confirmed` after test-run. The incident vocabulary is just the ordinary vocabulary.

is not; that limit is where the discussion begins.

7 Discussion and Limitations

Self-report honesty. The graph is only as truthful as its writer. The mitigations are structural: content hashes and commit bindings tie records to bytes, validation runs server-side in CI, the omit-don't-estimate rule removes the temptation of invented precision, and the human-only rungs cap what an AI may assert about its own work. The overwrite incident of Section 6 shows the mitigation working: with the repository as the process substrate, an AI mistake was a recoverable, recorded event rather than silent corruption.

From honesty to evidence: the dual-witness ask. The remaining weakness is that the record has one witness. Telemetry is provider-generated but agent-relayed: nothing stops a dishonest agent from underreporting usage, omitting sessions, or inventing activities. Provider-side provenance, signed inference records carrying model, timestamps, usage, roles, and prompt/response hashes, would eliminate that class of misreporting and subsume this paper's other provider asks (cost, energy, unambiguous roles). It would not, however, be tamper-*proof*, on three grounds. Trust shifts rather than vanishes: a signature proves the provider asserted the record, so key custody moves to the centre. Coverage stops at the API seam: the server can attest what passed through it, not which files were touched or how an attested inference binds to a repository artefact, so a dishonest client can still mis-bind. And judgements stay judgements: a signature proves a model emitted a verdict, not that the verdict is sound, and no server can attest that a human read a source. The realistic endpoint is a *dual witness*: provider-signed inference logs and repository-side commit binding, each referencing the other's hashes, so that falsifying the record requires collusion between independent parties with separately published histories, the same trust structure as certificate transparency or double-entry bookkeeping. A third witness is nearly free: anchoring each release's checksums in an append-only public log (certificate-transparency style, or a distributed ledger) makes retroactive rewriting of the repository history detectable without trusting

Table 3: Division of work in the meta-experiment, counted from the provenance graph and the session record at the time of writing (regenerated before submission).

Contribution	Agent	Value
Direction messages: scope, lineage, people, contested semantics (two ladder redesigns, cost policy, license, page budget)	human	141
Typed direction volume (median 7 words/message)	human	2 112 w
Active session time (request-gap clustering)	both	12.7 h
Authoring passes: prose, code, figures, method releases	AI	67
Audit passes: validation, promotions, content checks, incident reports, measurements	AI	163
Repair passes	AI	12
Deterministic builds logged	CI	2
Rungs granted above the AI ceiling	human	34

its host, though anchoring, like signing, certifies existence, never truth. aiprov supplies one witness and the seam for the others; signed provenance is the strongest standing ask this paper directs at providers.

Why human-only rungs stay human. `ai-confirmed` is a machine judgement with known failure modes (Alkaissi and McFarlane, 2023). The ladder does not pretend otherwise; it makes the judgement's author explicit instead of laundering machine confidence into apparent human endorsement. The human rung is no formality either: even domain experts struggle to override erroneous AI judgements against contradicting evidence (Vinas et al., 2026); hence the rung-4 gate hands the operator source and citing sentence, evidence before judgement.

Cost, and a gap that half closed. The method's overhead in this experiment was one logging command and one extra commit per working round, plus the audit activities the tooling writes itself. Because the session record carries per-request usage and tool calls, that overhead is measurable rather than anecdotal: a mechanical classification of every provider request by its tool invocations, bookkeeping commands (logging, validation, promotion, graph and transcript commits) versus everything else, attributes 12.6% of requests, 11.8% of output tokens, and 9.0% of the computed session cost to the method itself; on the repository side, 87 of 222 commits carried only graph or transcript updates. No request mixed bookkeeping with substantive work, so upper and lower bounds coincide: roughly a tenth of the activity is the price of making the other nine tenths auditable (Figure 8).

Token telemetry illustrates the omit-don't-estimate rule in motion: the harness exposes no live usage to the agent, so nothing was recorded at first; when the operator challenged the gap, the session record turned out to carry the provider's per-request usage blocks, and the totals in Table 2 were backfilled from them, reported rather than estimated. Cost closed the same way, one step later and one grade weaker: the record is unpriced, so under omit-don't-estimate no cost was recorded until the operator directed pricing the recorded tokens against the provider's published price list, first roughly USD 415, since

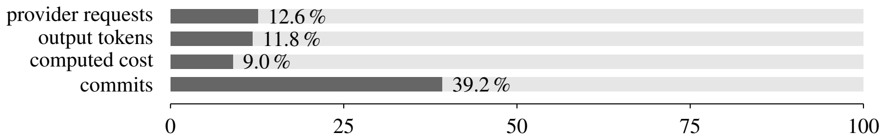

Figure 8: Method overhead as shares of the recorded session and repository, computed by the request classification described in the text and drawn from the metrics single source. The commits bar counts bookkeeping-only commits; every logging round commits, however small, hence its larger share.

refreshed in Table 2. The graph preserves the distinction between *reported* telemetry (token counts from the provider's usage blocks) and *computed* telemetry (cost, derived from them at a disclosed price list). Computed values are honest only while their basis is recorded; provider-reported cost would still be the stronger record. A larger question stays open behind the numbers: this one session consumed 860 million cache-read tokens, USD 1641 in all at list prices, for one draft paper and one method release, and whether that justifies its environmental and other costs is a debate this paper does not settle and does not hide. The schema reserves `aiprov:energyWh` for exactly this reason; it sits empty because no provider reported energy. Our position is only that the debate belongs over recorded consumption, not guesses, in either direction: neither dismissing AI assistance as obviously wasteful nor excusing it as obviously worthwhile. The graph makes the bill legible; whether it is worth paying is a judgement the record enables but cannot make.

Skill drift. The method evolved mid-experiment, v0 to v0.21, including two operator-contested ladder redesigns. Immutable archived snapshots, a changelog, and per-activity commit binding keep every run attributable to the exact method version in force, so evolution does not blur attribution.

Practices that emerged. Four, each small enough to adopt tomorrow; the first two are known from Manubot's continuous manuscripts (Himmelstein et al., 2019), rediscovered here in an agent-operated setting. Registry-fetched citation metadata: every BibTeX entry comes from doi.org content negotiation, never typed from memory, a concrete discipline against fabricated references (Walters and Wilder, 2023); the one failure was a registry serving malformed BibTeX, and CI caught it, not a reader. Registry-resolved identities: people enter the record via ORCID lookups, extending never-fabricate to social metadata. Verified-before-cited: no reference without a ladder-backed source node, making the bibliography a view of the graph. And content-checked before load-bearing, born from this paper's own audit (Section 6): any citation an argument leans on should reach `ai-confirmed` at minimum, with the checked depth disclosed in the promotion note.

Transparent self-citation. No-parentless-claim forces citing one's own priors; the lineage of Section 2 is the author's own repositories and specifications. Self-citation is frowned upon as metric gaming, yet unavoidable as provenance. The resolution is transparency instead of omission: sources carry `aiprov:selfCitation`, the report prints the ratio (here 5 of 75), and a reviewer can judge necessity per source; all five are operator-confirmed, the validator's link-only warning kept visible. The test that separates influence from inflation: a transparent self-citation is load-bearing, removing it would orphan a lineage or method claim, and it points at repositories and specifications rather than metric-bearing venues. It declares descent; it does not farm citations.

Regulatory alignment. The EU AI Act's transparency rules ask for disclosure and machine-readable marking of AI-generated content (European Parliament and Council of the European Union, 2024). A provenance graph is exactly that marking, and because CI derives the disclosure statement from the graph on every build, the published statement cannot lag the record. Beyond the EU, the OECD AI Principles, the first intergovernmental AI standard, ask AI actors to "ensure traceability, including in relation to datasets, processes and decisions made during the AI system lifecycle" (OECD, 2024): a per-activity provenance graph in all but name, and the OECD.AI observatory's incident monitoring (OECD.AI Policy Observatory, 2024) aggregates exactly such per-project records. Provenance-first authoring turns a compliance duty into a by-product.

Generality. Beyond papers, the same invariants and ladder apply to code, data pipelines, CAD models, and AAS submodels (Plattform Industrie 4.0, 2022); what remains domain work is the choice of entity subclasses. Generality also holds across clients and models: the skill bundle follows the open Agent Skills format (Anthropic, 2025), which independent agent clients implement, so the same bundle loads beyond the environment that produced this paper, and the graph is provider-agnostic, since any AI system, hosted or local, registers as an `AIAgent` with model and provider as data. Only three capture shims are client-specific (turn-end hook, transcript export, usage-field mapping); graph, ladder, and validator are plain Python and git. The AAS case bridges to the outlook: digital-twin infrastructures already assume machine-readable semantics, so AI-work provenance slots in as one more submodel concern rather than a foreign add-on.

Industry relevance: when the AI's output is a decision. The stakes sharpen outside publishing. In manufacturing, AI increasingly prepares or makes operational decisions, releasing a weld, classifying a part, adjusting a process window, and the worked example of Section 5 is one step from that setting. The same graph then answers what certification, liability, and the EU AI Act ask of high-risk systems, whose events must be automatically logged over the system's lifetime and which must remain under effective human oversight (European Parliament and Council of the European Union, 2024): which agent decided, on which evidence, under which model version, and who signed off. Aviation's certification guidance already anticipates exactly this setting: EASA's AI concept paper covers systems that automatically take decisions under human oversight and demands traceability of data from origin to final operation through the whole pipeline, naming record-keeping of AI-related data among the required methodologies (European Union Aviation Safety Agency, 2024). The human-only rungs map directly onto release decisions that must stay with people. The shepard platform does not yet support aiprov; the integration planned in Section 8 would carry the record into the same backbone that already holds the production data.

Originality of this work, on both sides of the division. Section 3 demands that novelty claims be bounded and checkable; the demand applies to this paper first. So the claim is made by the method's own mechanism: a logged novelty sweep refreshed the search horizon, its nearest find is named and cited, and the recorded claim is deliberately narrow. Per-activity AI provenance on a PROV substrate is *not* new, PROV-AGENT does that (Souza et al., 2025); what no found work combines is human-only verification rungs, executable-skill packaging, and self-application, and that combination, within the recorded horizon, is this paper's originality claim, refutable by widening the sweep. The effort behind it divides tellingly. The fast side, the AI's 12.7 active hours, supplied execution and recombination, and recombination is exactly what the empirical evidence says models supply: human-written texts contribute more new ideas to the collective pool than model-written ones (Moon et al., 2024). The slow side is where the originality entered: years of experiments, specification co-authoring, committee work, and conversations, the lineage of Section 2, compressed into 2 112 words of direction. Neither side alone would have produced the work: without the slow ideas the fast hours had nothing original to execute; without the fast execution the slow ideas would still be scattered across repositories and conversations. The record keeps both sides visible, which is precisely what an originality dispute would need.

Copyright when the machine wrote most of it. The repository's code is Apache-2.0-licensed, but a license can only grant what copyright first protects, and most of the repository's code and prose was AI-generated under human direction. The U.S. Copyright Office holds that copyright "does not extend to purely AI-generated material, or material where there is insufficient human control over the expressive elements", that prompts alone do not provide that control, and that protection attaches to human-authored expression and to "the creative selection, coordination, or arrangement" or modification of outputs (U.S. Copyright Office, 2025). German law agrees from its definition of the work: § 2(2) UrhG protects only *persönliche geistige Schöpfungen* (Bundesrepublik Deutschland, 1965), so current LLM output as such is not copyrightable in Germany. Read against this paper's own record, the conclusion is uncomfortable and clarifying at once: line by line, much of the artefact may be uncopyrightable, and the license's effective scope narrows to the human contributions and the curated whole. What the method adds is the evidence a legal answer needs: which expression a human conceived, selected, arranged, or modified, and which the machine produced under how much control, is exactly what the per-activity graph records (Table 3) and what a blanket "AI-assisted" disclaimer erases. Rights analysis of AI-era works founders on evidence before doctrine; provenance-first authoring is how the evidence survives.

Nostalgia for the paper, or the wrong artefact instrumented? The sharpest objection to this work is not about any rung but about its target: perhaps F(AI)^{2}R merely patches provenance onto a form we are nostalgic for, while AI-driven science becomes something structurally different: living claim graphs, continuously re-executed analyses, agents exchanging machine-readable evidence, narrative generated on demand. If so, instrumenting PDF production is polishing the wrong artefact. But the method is less paper-bound than its demonstration: the invariants and ladder attach to claims, sources, activities, and agents, never to a manuscript, and here the graph is the primary record, the PDF one derived view CI rebuilds from it. Read that way, F(AI)^{2}R is a migration path, moving a publication's substance into the claim-level form a post-paper science would need while still emitting the legacy format. If the paper form dissolves, judge this method by whether its graphs remain useful without the PDFs.

The force multiplier, both ways, and the optimistic reading. One operator ran method, paper, and integration fork in parallel during this experiment; unaccounted, the same multiplication is the dilution engine of Section 3. The differentiator is not speed but auditability: fast work that carries its record is distinguishable from fast noise. Fast paper, slow ideas is the honest reading of this project, and only the graph makes that reading checkable. There is also a hopeful version of the same observation. What the multiplication buys is not mainly volume: AI absorbing the tedium (registry lookups, hash binding, conformance checks, build plumbing) can free human attention for the two things machines do not supply, connecting ideas and connecting people. This experiment's own activity log shows that division of labour. No guarantee, only a chance; a provenance graph is how one would check, later, whether the chance was taken. For this experiment the leverage measurement of Section 6 is that check, taken early: 2 112 typed words steering 12.7 active hours, nearly all of them spent on scope, semantics, lineage, or people.

8 Conclusion

We generalized the F(AI)^{2}R method into aiprov, a domain-agnostic PROV-O extension for AI-in-the-loop work; we packaged the method as an executable skill an AI agent operates itself; and we applied it to its own production, mishaps, mid-experiment redesigns, and all. The stance of the original experiment carries over and extends: the repository is the paper, is the process, and now also is the method, versioned, validated, and shipped alongside what it produced.

Three roots converge here: AI-assisted research practice, which contributed transcript-as-artifact; engineering provenance, which insists on traceability; and industrial semantic modeling, which taught that where exchange matters, semantics must be formal. Where semantics are important, provenance is the semantics of work.

The path forward is infrastructure. shepard (Haase et al., 2025), a jointly developed DLR system for heterogeneous product and research data, already serves as the data backbone of engineering deployments such as the MEMAS additive-manufacturing pipeline (Unger et al., 2024); an experimental fork (Krebs, n.d.c) is the staging ground for integrating aiprov into it as planned work. Where shepard instances capture what was produced and measured, aiprov adds who, human or AI, shaped it and how it was verified. Further work includes multi-agent attribution, energy accounting, registry integration beyond ORCID and DOI, vendoring sources at the access gate by default, a combined profile with uncertainty-aware provenance (Valente et al., 2026), and registering the vocabulary namespace as a permanent identifier, deliberately deferred so that community feedback shapes the terms before the identifier freezes. Two further items come straight from reviewing this work against its own standard. The `ai-confirmed` judgements here were granted by the agent that wrote the citing sentences; measuring that self-confirmation bias, by having an independent model or sampled humans re-verify the same sources, is the next audit the method owes itself. And the meta-experiment is a single self-applied case; the portability claims of Section 7 earn their keep only when a second operator applies the skill in a second domain and the records are compared.

One thing must be said with urgency, and plainly. Nothing in this paper settles what science with AI in the loop should become. That understanding is being negotiated now, by editors rewriting authorship policies, regulators phasing in transparency duties, funders and communities arguing over credit, originality, cost, and trust, and it will be settled socially, not technically. This initiative is not the solution; it is technical support for whichever understanding society arrives at, a way to keep the record straight while the norms are still moving. But the discussion itself cannot wait, because infrastructure that ships first tends to settle norms by default, and the most expensive mistake would be to let tools decide what the community has not yet debated. We ask for that debate, and we offer this repository's record, its graph, its transcript, its incidents, its bill, as material evidence to hold it over.

Acknowledgment and AI Transparency Statement

The ideas in this paper were sharpened through earlier experiments (Krebs, n.d.b,n); through discussions in the context of the Helmholtz Metadata Collaboration (HMC) (Helmholtz Association, n.d.):

with Witold Arndt in the HMC hub for aeronautics, space and transport, and through the broad view of AI in research afforded by chairing the HMC Conference 2025 programme committee (Helmholtz Metadata Collaboration, 2025); through exchanges with Frank Dressel on provenance for engineering systems (Valente et al., 2026); through deep discussions with Sirko Schindler and Carsten Hoyer-Klick on ontologies; through the research-data-management evangelism of Christian Langenbach at DLR; and through the inspiration of Carina Haupt's work on provenance use cases (Haupt, 2022) and of the MEMAS project's integrated data management (Unger et al., 2024).

This paper and its repository were produced with an AI agent operating the ai-provenance skill; every activity, generated artefact, claim, and source is recorded in the version-controlled provenance graph, including the production of this statement. The following disclosure is generated from that graph by `provlog.py disclosure` and is made in view of the transparency rules of the EU Artificial Intelligence Act (European Parliament and Council of the European Union, 2024):

Parts of this work were generated or edited with the assistance of artificial-intelligence systems: Claude Code (Claude Fable 5, Anthropic), associated with 206 recorded activities. In line with the transparency obligations for AI-generated content under Regulation (EU) 2024/1689 (AI Act), this assistance is disclosed here and is additionally marked in machine-readable form: the version-controlled provenance graph (provenance.ttl, W3C PROV-O) records the 244 activities behind this work, the 61 generated artefacts with content hashes, and the 8 recorded claims with their verification states. Human oversight is structural: verification rungs above ai-confirmed are reserved to human agents, and the conformance validator rejects AI-granted promotions.

Author contributions (CRediT): Florian Krebs: conceptualization, methodology, investigation, validation, data curation, writing (original draft), writing (review and editing), supervision. All roles were exercised with AI assistance operating under the author's direction, as disclosed in the transparency statement above.

Competing interests: the author declares none.

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Data availability: all data essential for reproducing the findings, that is, the machine-readable provenance graph (`provenance.ttl`), the computed metrics (`metrics.json`), the per-request telemetry export, the source verification worksheet, and the full session transcripts, are openly available in the version-pinned release v0.17.0 of the repository at `https://github.com/noheton/f-ai2-r` and are archived on Zenodo under a persistent identifier (Krebs, 2026). The skill bundle and all analysis code are available in the same release; the skill loads in any Agent Skills client, and conformance-gated releases carry bundle, PDF, graph, and checksums. A preprint of this article is available on arXiv.

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